Time left 0:43:58

Question 6

Not yet answered

Marked out of 12.00

1. Figure 1 below illustrates the operation of channel I/O. Suppose the system has a request to transfer data from Tape X to Printer Y, analyze the figure below and formulate the path of communication of the transfer operation. Assume that CPU has placed the I/O program in the memory and the IOP is ready to fetch the instruction.

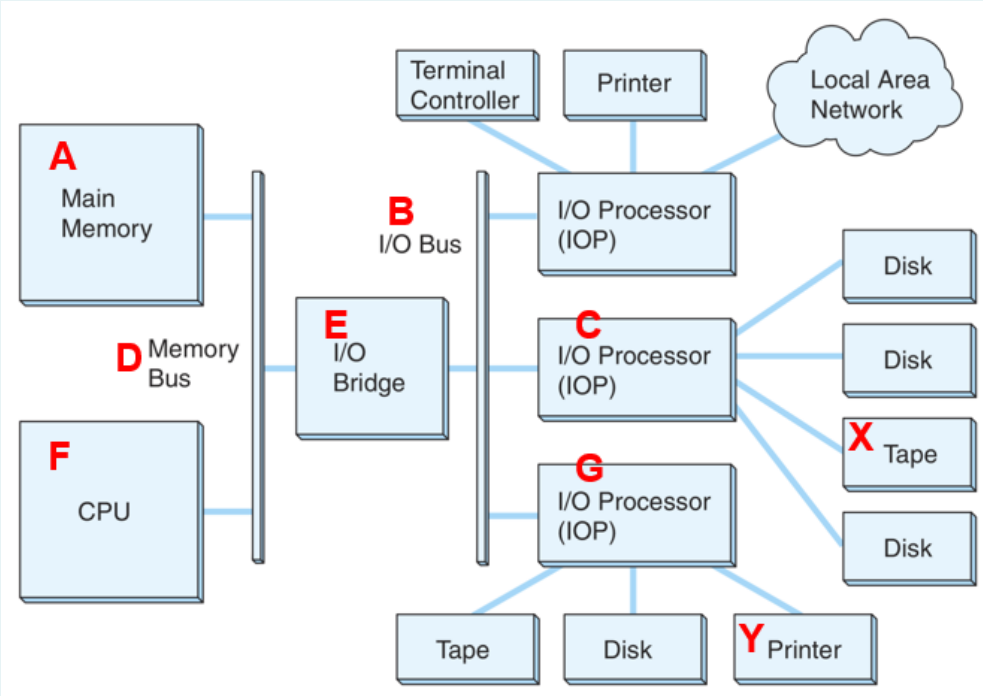


Figure 1: Channel I/O

Answer format: Write the path of communication using only the label in red separated by a comma with no white space in between, e.g. Printer Y communicating with I/O bus through IOP is written as Y,G,B

IOP fetches instruction for transfer operation of Tape X

IOP execute the transfer to Printer Y

2. Consider a server with RAID Level 3 and given Figure 2 below, determine the bit for data regeneration.

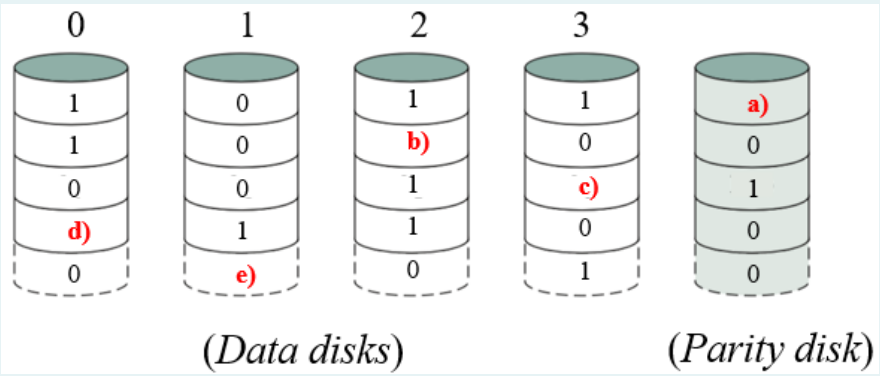


Figure 2: Server with RAID 3

a)

b)

c)

d)

e)

3. Suppose there are 6 data drives and they need to be stored as RAID 5, calculate how many drives are needed altogether.

The drives required is determined by with N is the . Therefore,

drives are needed.

4. Suppose a system has 5 drives as in Figure 3. The system needs to be configured as RAID 6. Given the block of data in the Figure 4 below, design them to allow dual redundancy.

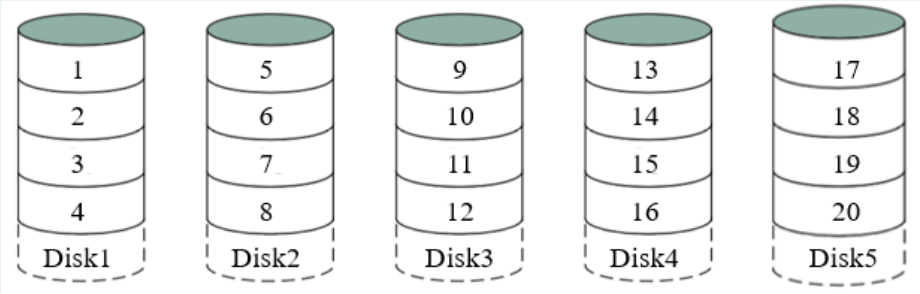


Figure 3: System with 5 disk drive

The number in the disks are the location. E.g. number 5 is referring to location 5

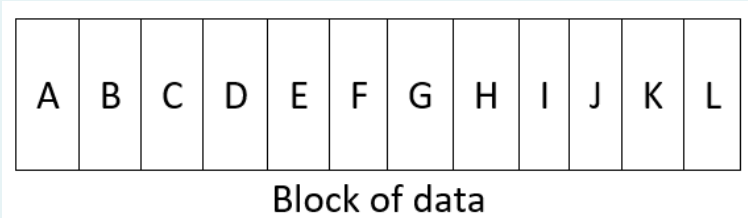


Figure 4: Block of data

a) Determine which block of data are allocated at location 3, 15 and 17
Answer format: Write only the letter with no white space. E.g. at location 1, A is written to represent data block A. Use uppercase letter P to represent parity block.

Location 3 =

Location 15 =

Location 17 =

b) If Disk1 and Disk4 failed, determine which location(s) should be used to regenerate the data at location 2.

Answer format: Write only the letter with no white space. If your answer has more than one location separate the letters with a comma e.g. location 1 and location 2 are written as 1,2 with a comma with no white space

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Data retention summary

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